

Demonstration of Proposed Features

Date	03 July 2026
Team ID	XXXXXX
Project Name	Voice-Based Concept Understanding Analyser
Maximum Marks	1 Mark

Demonstration of Proposed Features:

This document captures all the features that were proposed during the project planning phase and tracks whether each feature was successfully implemented and demonstrated. It serves as evidence of the team's ability to deliver on the proposed solution.

S.No	Feature Name	Description	Status (Implemented / Partial / Pending)	Demonstrated (Yes / No)	Remarks
1	Audio Upload & Format Handling	Accepts student audio file uploads (MP4/WAV) and converts them using FFmpeg.	Implemented	Yes	Successfully handles multiple audio streams
2	Speech-to-Text Transcription	Transcribes audio waveforms into textual content automatically.	Implemented	Yes	Accurate text extraction using speech recognition libraries
3	Conceptual Evaluation Engine	Uses Gemini 2.5-flash to assess technical comprehension and generate scores (0–100).	Implemented	Yes	Generates structured evaluations based on conceptual accuracy.
4	Fluency & Filler Analytics	Identifies and logs specific verbal filler frequencies (e.g., um, uh, like, ah).	Implemented	Yes	Parsed accurately via strict structured JSON parameters.
5	Real-time Speech Pacing Metrics	Calculates runtime session duration, exact word counts, and words per minute (WPM).	Implemented	Yes	Metrics are calculated instantly upon file upload processing.
6	Automated PDF Reports	Compiles complete individual evaluation logs into formal, downloadable PDF files.	Implemented	Yes	Generates clean stream outputs via the ReportLab toolkit library.

Feature Implementation Summary:

Total Features Proposed	6
Total Features Implemented	6
Total Features Demonstrated	6
Overall Implementation Rate (%)	100%